

SUPPLY CHAIN ANALYSIS

Bristol One City: Urban Delivery Chain

7-phase delivery chain analysis. Chain balance 81.3% (Good). Primary bottleneck: Phase 5 Community and Household Uptake (58.3%, Poor). The city has exemplary institutional infrastructure but a critical disconnect at the point of contact with deprived communities.

81.3%

Chain average balance (Good)

194

Average N score (chain coherence)

2

Bottleneck phases (P5, P3)

100 pts

D-C gap at Phase 5 (largest in chain)

ANALYSIS PAGES



N-D-C Dashboard

Phase-level D, C, N bars; coherence pentagon; bottleneck alert; full phase data table.



Project Context

Geographic and governance scope; bond opportunity analysis; Phase Intervention Worksheet; data sources.



Just Transition Return Chain

TVPCI-R community wealth recirculation analysis. TVPCI-R = 50.7 (Failure). B_chain = 47.0.

PHASE MAP: 7-PHASE URBAN DELIVERY CHAIN

PHASE	NAME	D	C	BALANCE	N	STATUS	TRANSPARENCY	KEY CONSTRAINT OR OUTPUT
P0	Vision and Mandate	270	240	88.9%	226	GOOD	HIGH	5 plan iterations; 2030 net-zero; 4 missions; 17 SDGs; BACCC
P1	Governance and Coordination	250	210	84.0%	192	GOOD	MEDIUM	7 thematic boards; Governance

PHASE	NAME	D	C	BALANCE	N	STATUS	TRANSPARENCY	KEY CONSTRAINT OR OUTPUT
								Board (12 members); BACCC independent review
P2	Strategy and Program Design	265	240	90.6%	228	EXCELLENT	HIGH	City Leap binding KPIs; IBP £631m; BCAI Green Finance Framework; LNRS
P3	Capital Deployment	275	195	70.9%	164	FAIR	MEDIUM	£1.67bn capital budget; BCAI only £2m raised; scope 3 capital near-zero
P4	Infrastructure Delivery	260	220	84.6%	202	GOOD	HIGH	City Leap KPIs on track; heat network; solar at schools; housing retrofit; 10% BNG
P5 	Community and Household Uptake	240	140	58.3%	107	POOR	LOW	750 BCAI investors from 470k population; 10.7yr life expectancy gap; scope 3 untracked
P6	Outcome Measurement	270	248	91.9%	238	EXCELLENT	HIGH	256 QoL indicators; PHOF; City Leap KPI audit; VLR; quarterly

PHASE	NAME	D	C	BALANCE	N	STATUS	TRANSPARENCY	KEY CONSTRAINT OR OUTPUT
								Power BI dashboard

N-D-C FRAMEWORK APPLIED TO URBAN POLICY DELIVERY

In the Tholonic N-D-C framework, **D (Definition)** captures the constraints, requirements, governance obligations, and binding commitments that define what each delivery phase must achieve. **C (Contribution)** captures the actual outputs, measurable delivery, and flows produced by each phase. **N (Negotiation)** is the coherent emergent state of the phase: the product of D and C in balance.

For Bristol, D across Phases 0-2 is very high: the city has invested heavily in policy commitments, governance structures, and binding strategies. The D-C gap widens at Phase 3 (capital) and collapses into a structural failure at Phase 5 (community uptake). Phase 6 (measurement) recovers because Bristol's monitoring infrastructure is genuinely world-class: the QoL survey (256 indicators), BACCC independent scrutiny, and City Leap contractual audit form an unusually robust verification layer. This measurement strength is the foundation on which a Sustainability-Linked Bond KPI framework can be built.